

CHE 253K

Applied Statistics

Spring 2026 (16520)

Course Description: The primary aims of this course are to understand how to apply statistics to learn from data and apply that knowledge to make decisions. Statistics and the larger growing field of data analytics are extensive. Here you will learn basic concepts and techniques of data analysis which form the foundation of these subjects. You will also gain experience managing and analyzing datasets using Excel and JMP.

Prerequisites: ChE 210 (Introduction to Computing), and M 427J (Differential Equations). **Co-requisite:** ChE 317 (Introduction to Chemical Engineering Analysis).

Class Meeting: Lectures: MW 9:00-10:00am, PMA 5.104
Recitation: F 9:00-10:00am, PMA 5.104

Instructor: Dr. David T. Dalle Molle (512-471-7722 | dtd@che.utexas.edu)
Office Hours: Mondays 1-2pm and Wednesdays 11am–noon in CPE 3.458 or by appointment

Evaluation Method:	In class Concept Quizzes	10%
	Homework	15%
	Mid-Term (2) and Final Exams	75%
	TOTAL	100%

Mid-Term Exam 1: Wednesday, February 25, 2026, 7:00-9:00pm – *proposed**

Mid-Term Exam 2: Wednesday, April 8, 2026, 7:00-9:00pm – *proposed**

Final Exam Date: Saturday May 2, 2026, 3:30pm – 5:30pm – *set by UT***

**Exam dates will be finalized in the 1st week of class once any known conflicts have been resolved.*

***Tentative – Final Exams Schedule are confirmed on the 20th class day.*

Note that grades will be curved as necessary and letter grades assigned based on natural breaks in the distribution. Plus/Minus grading will be used in this course.

Required Text: *Probability and Statistics for Engineering and the Sciences*, 9th ed. Jay L. Devore, Cengage Learning, Boston, MA. (ISBN: 9781305251809)

NOTE: The materials for this class are available through the Longhorn Textbook Access (LTA) program, an initiative between UT Austin, The University Co-op and textbook publishers to significantly reduce the cost of digital course materials for students. You are automatically opted into the program but can easily opt-out (and back in) via Canvas through the 12th class day. To opt out, simply press the “My Textbooks” link in Canvas and then press the “Opt Out” button on the material. You will be billed

for your LTA material after the 5th class day through your ‘What I Owe’ page but still have through the 12th class day to opt out. You must pay for your LTA materials through the What I Owe page by the 20th class day. If you do not pay by the 20th class day, you will lose access to the materials after the 20th class day, and your charge will be removed. More information about the LTA program is available at [Longhorn Textbook Access – University Co-op](#). Information on the VitalSource off-line reader can be found [here](#).

Course Topics: The course is composed of 28 50-minute lectures as outlined below.

Lecture No.	Topic
1	Intro, course objectives, syllabus review, basics
2-3	Descriptive statistics
4	Basic Rules of Probability
5	Solving probability problems with tree diagrams
6	Conditional probabilities and Bayes Theorem
7	Discrete probability distribution functions
8	Binomial distribution function and applications
9	Continuous probability distribution functions & The Normal Distr.
10	The t-distribution and its properties & the Central Limit Theorem
11	Combinations of Distributions
12-13	Confidence intervals
14-17	Hypothesis Testing
18-20	Statistical process control
21-24	Linear and non-linear regression
25-27	Design of Experiments/Response Surface Models
28	Course review

Note: Information posted on Announcements in Canvas ([Sp26 - APPLIED STATISTICS \(16520\)](#)) supersedes that in this syllabus. It is your obligation to read and note any changes in the Announcements.

Lectures: Lectures will be given in person at the scheduled lecture time. **You are expected to attend the lectures in person (see also the Important Note below on Concept Quizzes).** Lectures will not be broadcast by Zoom. There will be ample opportunity to ask questions during the lectures, recitation sessions, or at office hours.

Recitation Sessions: Recitation Sessions will be held in person at the scheduled recitation time. **You are expected to attend these sessions in person (see also the Important Note below on Concept Quizzes).** Recitation Sessions will not be broadcast by Zoom. There will be ample opportunity to ask questions about lecture concepts, homework, or exams.

Concept Quizzes: At some time in some lectures or some recitation sessions, you will be given a short concept quiz using the UT Instapoll tool in Canvas. Each quiz will consist of a single question. A correct answer is worth 1 point. An incorrect answer is worth 0.5 points. If you do not take the quiz, your score is 0 points. There will likely be between

15-20 Concept Quizzes. The two lowest quiz grades will be dropped. The purpose of these concept quizzes is to help you to stay engaged and master the material.

Important Note: ***Concept quizzes may be given at any time during a lecture period and will be given in most Recitation sessions. You are expected to take Concept Quizzes even if you are not attending class in person that day. UT Instapoll is available through Canvas either using your laptop browser or via the Canvas app on your phone, so you may take the quiz even if you are not at the class in person.***

Homework: Problem sets will be issued roughly once a week (10 or 11 assignments for the semester). It is essential that you complete the problem sets on time because each lecture and homework build on previous homework. A significant part of your mastery of the subject will come from you working out the solutions in detail. The TAs and I are available during office hours and by appointment to help you. There may also be tutors available in CPE to help around the homework due dates. More on this matter in class.

You must follow the homework submissions guidance and sample that are posted in Canvas. You are expected to have made a genuine attempt at the problems in question before seeing us with questions because it will help you in mastering the material. Late homework will be counted as 25% of the total available points for each day that it is late. You are allowed to drop your lowest Homework assignment.

Exams: There are two midterm exams and one final exam. They are 2 hours long and equally weighted. There are no make-up exams.

Only scientific, non-graphing calculators can be used on exams. Calculators with CAS or graphing capabilities are not allowed. Please make arrangements to have an acceptable calculator for the exams. Unapproved calculators will be taken away from students if seen in use during an exam. If a correct answer is given on an exam and there is no supporting work, there will be no credit given for that answer. You must show all your work leading to the correct answer.

While taking mid-term and final exams, you may not use any lecture notes, homework solutions, notes from the lecture that I post on Canvas, or any other documents/textbooks unless specifically agreed to for a given exam. If it is discovered that you have violated this rule, you will receive a zero for the exam and you will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students ([Office of the Dean of Students - Student Conduct and Academic Integrity](#)). These reports can result in sanctions, including failure in the course.

Regrading: If you feel a homework or exam was not correctly or fairly graded, or that it was simply not totaled correctly, you may submit a regrade request that must include a specific explanation for the request relative to the published solution/rubric.

All regrade requests must be submitted through Gradescope within 48 hours of when the graded document is available for review to the class. Failure to promptly review your graded homework or graded exam soon after they are published in Gradescope does not entitle you to any additional time to make a regrade request. Note that it is possible that regrading could result in a lower grade should other issues come to light in the review.

Student Outcomes: This class will help provide Student Outcomes 1 and 6 of the Accreditation Board for Engineering and Technology (ABET):

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

Academic Accommodations: The University of Texas at Austin provides upon request appropriate academic accommodations for qualified students with disabilities. For more information, contact the [Division of Diversity and Community Engagement, Accommodations and Services](#) at access@austin.utexas.edu or 512-471-6259 (audio), 512-410-6644 (videophone).

Academic Integrity: Students who violate University rules on scholastic dishonesty (including cheating on tests, plagiarism on written assignments, collaboration on assignments that are specifically intended to be non-collaborative, etc.) are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. Since such dishonesty harms the individual, all students, and the integrity of the University, policies on scholastic dishonesty will be strictly enforced. It is each and every student's responsibility to be familiar with the University rules and guidelines on academic integrity and scholastic dishonesty, which are posted at [Office of the Dean of Students - Student Conduct and Academic Integrity - Standards of Conduct](#).

Use of Artificial Intelligence (AI) Tools: This course assumes that work submitted by students – all process work, drafts, brainstorming artifacts, final works – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment or exam, and/or using generative AI tools (such as ChatGPT).

In this course, every element of class assignments must be fully prepared by the student. The use of generative AI tools for any part of your work will be treated as plagiarism. If you have questions, please contact me.

All assignments should be fully prepared by the student. Developing strong competencies in the skills associated with this course, from student-based brainstorming to project development, will prepare you for success in your degree pathway and, ultimately, a competitive career. Therefore, the use of generative AI tools to complete any aspect of

assignments for this course is not permitted and will be treated as plagiarism. If you have questions about what constitutes a violation of this statement, please contact me.

This course assumes that work submitted for a grade by students – all process work, drafts, brainstorming artifacts, final works – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment or exam, and/or using generative AI tools (such as ChatGPT).

Title IX, a segment of the Education Amendments of 1972, makes it clear that violence, harassment, or discrimination based on sex and gender are Civil Rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, sexuality, disability, etc. If you or someone you know has been harassed or assaulted, please don't hesitate to speak to anyone on our teaching team. You can also find useful resources at [Support & Resources | Title IX Office](#).

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without my explicit, my written permission. Unauthorized sharing of materials may facilitate cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to [Student Conduct and Academic Integrity](#) in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

Class Recordings: Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Classroom Safety and COVID-19: To help preserve our in person learning environment, please utilize the resources available at the [Healthy Horns website](#).